Cleaning Robot Planning Report

This report summarizes the plan generated for the autonomous cleaning robot.

Goal

The goal is to clean both the classroom and the cafeteria, and to dispose of all trash. This means both rooms must be mopped, and all trash must be in the dustbin.

Plan

The generated plan is a sequence of 12 actions:

1. **Sweep and Mop Classroom:** The robot starts in the classroom, sweeps it, and then mops it.
2. **Collect Trash:** The robot picks up the trash in the classroom.
3. **Move to Cafeteria:** The robot moves to the cafeteria.
4. **Sweep and Mop Cafeteria:** The robot sweeps and mops the cafeteria.
5. **First Trip to Dustbin:** The robot moves to the dustbin to drop the trash from the classroom.
6. **Return to Cafeteria:** The robot goes back to the cafeteria to pick up the trash there.
7. **Second Trip to Dustbin:** The robot returns to the dustbin to dispose of the second piece of trash.

Conclusion

The plan is optimal as it minimizes the number of actions required to achieve the goal. The robot successfully cleans both rooms and disposes of all trash, fulfilling all goal conditions. The two trips to the dustbin are necessary because the robot can only carry one item of trash at a time.